CS & IT ENGINEERING

Digital Logic

Miscellaneous Topics

Lecture No. 1

By- Aditya sir

Recap of Previous Lecture

Topic

Synchronous Counters

→

Topics to be Covered

Topic

Number System

About Aditya Jain sir

1. Appeared for GATE during BTech and secured AIR 60 in GATE in very first attempt - City topper
2. Represented college as the first Google DSC Ambassador.
3. The only student from the batch to secure an internship at Amazon. (9+ CGPA)
4. Had offer from IIT Bombay and IISc Bangalore to join the Masters program
5. Joined IIT Bombay for my 2 year Masters program, specialization in Data Science
6. Published multiple research papers in well known conferences along with the team
7. Received the prestigious excellence in Research award from IIT Bombay for my Masters thesis
8. Completed my Masters with an overall GPA of 9.36/10
9. Joined Dream11 as a Data Scientist
10. Have mentored 12,000+ students & working professions in field of Data Science and Analytics
11. Have been mentoring & teaching GATE aspirants to secure a great rank in limited time
12. Have got around 27.5K followers on Linkedin where I share my insights and guide students and professionals.

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Topic : Number System

Base (Radix)

Total number of digit used in the system

- Number System
  - Decimal Number → Base -10 {Digit 0-9}
  - Binary Number → Base -2 {Digit 0, 1}
  - Octal Number → Base -8 {Digit 0-7}
  - Hexadecimal Number → Base -16 {Digit 0-9, A - F}

Topic : Number System

Base (Radix)

Decimal:

wt

.....	10^4	10^3	10^2	10^1	10^0	10^{-1}	10^{-2}	10^{-3}	.....
.....	a_4	a_3	a_2	a_1	a_0 .	a_{-1}	a_{-2}	a_{-3}	.....

a → Coefficient of decimal number system

10→ Weight of decimal number system

↓

importance

Example:

$(501.23)_{10}$

10^{2}	10^{1}	10^{0}		10^{-1}	10^{-2}
5	0	1	.	2	3

Weight → 10^{2} 10^{1} 10^{0} 10^{-1} 10^{-2}

Coefficient → $(\cancel{125.75})_{10}$

5 0 1 . 2 3

Topic : Number System

V. imp

Base	Digit
2	0, 1
3	0,1,2
4	0,1,2,3
5	0,1,2,3,4
6	0,1,2,3,4,5
7	0,1,2,3,4,5,6
8	0,1,2,3,4,5,6,7
9	0,1,2,3,4,5,6,7,8
10	0,1,2,3,4,5,6,7,8,9
11	0,1,2,3,4,5,6,7,8,9,A
12.	0,1,2,3,4,5,6,7,8,9,A,B
13.	0,1,2,3,4,5,6,7,8,9,A,B,C
14.	0,1,2,3,4,5,6,7,8,9,A,B, C,D
15.	0,1,2,3,4,5,6,7,8,9,A,B, C,D,E
16.	0,1,2,3,4,5,6,7,8,9,A,B, C,D,E,F

$A \rightarrow 10$
$B \rightarrow 11$
$C \rightarrow 12$
$D \rightarrow 13$
$E \rightarrow 14$
$F \rightarrow 15$

$0 \longrightarrow b-1$

(101)

2, 3, 4 . . . . . .

Binary Number System (Base (Radix) = 2]

.....	2^4	2^3	2^2	2^1	2^0	2^{-1}	2^{-2}	2^{-3}	.....
.....	a_4	a_3	a_2	a_1	a_0	a_{-1}	a_{-2}	a_{-3}	.....

$2^i \rightarrow$ Weight of Binary number system

$a_i \rightarrow$ Coefficient of Binary number system {0,1}

$0 \rightarrow b-1$

Example:

$(101.11)_2$ ← Binary

2^2	2^1	2^0		2^{-1}	2^{-2}
1	0	1	.	1	1

Octal Number System (Base (Radix) = 8J

.....	8^4	8^3	8^2	8^1	8^0	8^{-1}	8^{-2}	8^{-3}	.....
.....	a_4	a_3	a_2	a_1	a_0 .	a_{-1}	a_{-2}	a_{-3}	.....

8^i → Weight of ~~Binary~~ Octal number system

a_i →Coefficient of Binary number system {0,7}

Topic : Number System

Example:

$(720.64)_8$

8^4	8^1	8^0		8^{-1}	8^{-2}
7	2	0	.	6	4

Topic : Number System

Hexadecimal Number System (Base (Radix) = 16)

.....	16^4	16^3	16^2	16^1	16^0	16^{-1}	16^{-2}	16^{-3}	.....
.....	a_4	a_3	a_2	a_1	a_0 .	a_{-1}	a_{-2}	a_{-3}	.....

$16^i \rightarrow$ Weight of ~~Binary~~ Hexadecimal number system

$a_i \rightarrow$ Coefficient of Binary number system {0 –9, A – F}

10 – 15

$$(A2C.F)_{16}$$

Topic : Number System

Example:

$(A2C.F)_{16}$

16^2	16^1	16^0		16^{-1}
A	2	C	.	F

In base conversion 2 key points are there:

(A) Any base to Decimal conversion

(B) Decimal to any other base conversion

} Conversion

(Q.1) $(n)_b \longrightarrow (?)_{10}$

(Q.2) $(n)_{10} \longrightarrow (?)_b$

(Q.3) $(n)_{b1} \longrightarrow (\ )_{b2}$

$(n)_{b1} \rightarrow (\ )_{10} \rightarrow (\ )_{b2}$

Topic : Number System

In base conversion 2 key points are there:

$(\ )_r \longrightarrow (\ )_{10}$

(A) Any base to Decimal conversion

$$(\overset{r^3}{a_3} \quad \overset{r^2}{a_2} \quad \overset{r^1}{a_1} \quad \overset{r^0}{a_0} \, . \, \overset{r^{-1}}{a_{-1}} \quad \overset{r^{-2}}{a_{-2}})_r = (\ )_{10}$$

$$(a_3 \times r^3 + a_2 \times r^2 + a_1 \times r^1 + a_0 \times r^0 + a_{-1} \times r^{-1} + a_{-2} \times r^{-2}) = (\ )_{10}$$

$$(1011.11)_2 \longrightarrow (\,?\,)_{10}$$

Topic : Number System

Case-1: Binary to Decimal conversion

$2^3\ 2^2\ 2^1\ 2^0\ 2^{-1}\ 2^{-2}$

Example: $(1011.11)_2 = (\ )_{10}$

$\Rightarrow [(1\times2^3) + (1\times2^2) + (1\times2^1) + (1\times2^0) + (1\times2^{-1}) + (1\times2^{-2})]_{10}$

$[8 + 0 + 2 + 1 + 0.5 + 0.25]_{10}$

$\Rightarrow \boxed{(11.75)_{10}}$

010

$(729.4)_8 \rightarrow (?)_{10}$

invalid

$(721.4)_8 \rightarrow (?)_{10}$

Topic : Number System

Case-2: Octal to Decimal conversion

$8^2 \; 8^1 \; 8^0 \; 8^{-1}$

Example: $(721.4)_2 = (\;)_{10}$

$\Rightarrow [(7\times8^2) + (2\times8^1) + (1\times8^0) + (4\times8^{-1})]_{10}$

$\Rightarrow [448 + 16 + 1 + 0.5]_{10}$

$\Rightarrow (465.5)_{10}$

$$(A2B.C)_{16} \longrightarrow (\,?\,)_{10}$$

Case-3: Hexadecimal to Decimal conversion

$16^2\ 16^1\ 16^0\ 16^{-1}$

Example: $(A2B.C)_{16} = (\,)_{16}$

$\Rightarrow [(A \times 16^2) + (2 \times 16^1) + (B \times 16^0) + (C \times 16^{-1})]_{10}$

$\Rightarrow [(10 \times 256) + (2 \times 16) + (11 \times 1) + (12 \times 16^{-1})]_{10}$

$\Rightarrow [2560 + 32 + 11 + 0.75]_{10}$

$\Rightarrow (2603.75)_{10}$

$\boxed{A \rightarrow 10}$

$$(432.22)_5 \longrightarrow (\,?\,)_{10}$$

Topic : Number System

Case-4: Base 5 to Decimal conversion

Example: $(432.22)_5 = (\ )_{10}$

$\Rightarrow [(4 \times 5^2)+(3 \times 5^1)+(2 \times 5^0)+(2 \times 5^{-1})+(2 \times 5^{-2})+]_{10}$

$\Rightarrow [100 + 15 + 2 + 0.4 + 0.08]_{10}$

$\Rightarrow (117.48)_{10}$

$$(4365.23)_7 \longrightarrow (\ ?\ )_{10}$$

[NAT]

$3 \quad 2 \quad 1 \quad 7^{0} \quad -1 \quad -2$

#Q. $(4365.23)_7 = (?)_{10}$

$$\Rightarrow [(4\times7^3) + (3\times7^2) + (6\times7^1) + (5\times7^0) + (2\times7^{-1}) + (3\times7^{-2})]_{10}$$

$$\Rightarrow (1566.346)_{10}$$

[NAT]

Tricky

#Q. $(62C) = (?)_{10}$

By minimum base convert into decimal?

37.5%

$(62C)$ ✓ 13, 14, 15 . . .

$A \rightarrow 10$

$B \rightarrow 11$

$\boxed{C \rightarrow 12}$

base = 12

6 2 C

13^2 13^1 13^0

$6 \times 13^2 + 2 \times 13^1 + 12 \times 1 = (1052)_{10}$

(2) Decimal $\longrightarrow$ Any base

$$(n)_{10} \longrightarrow (?)_b$$

In base conversion 2 key points are there:

(B) Decimal to any other base conversion

$(n)_{10} \longrightarrow (?)_r$

$n \div r$

coef rem

r	a_3	a_2	a_1	a_0
r		coeff		
r		( )		

Remainder

b_0

b_1

b_2

b_3

$(\underbrace{a_3a_2a_1a_0}.a_{-1}a_{-2}a_{-3})_{10}$

[redacted]

Topic : Number System

0.75

$0.\ a_{-1}\ a_{-2}\ a_{-3} \times r = X_0 \cdot X_{-1} X_{-2}$

$0.\ x_{-1}\ x_{-2} \times r = X_1 \cdot X_{-3} X_{-4}$

$0.\ x_{-3}\ x_{-4} \times r = X_2 \cdot X_{-5} X_{-6}$

$(a_3 a_2 a_1 a_0 \cdot a_{-1}\ a_{-2}\ a_{-3})_{10} = (b_3 b_2 b_1 b_0 \cdot x_0 x_1 x_2)_r$

Final

eg

$(723.25)_{10} \longrightarrow (\quad)_3$

$0.25 \times 3 = 0.75$

$0.75 \times 3 = 2.25$

0.25×3

Ans

$.02$

① eg:- $(19.75)_{10} \longrightarrow (?)_2$

Telegram

Case-1: Decimal to Binary Base conversion.

Example: $(19.75)_{10} = (\ )_2$

Before Decimal

After Decimal

		Rem
2	19	1
2	9	1
2	4	0
2	2	0
	1	1
	0	Stop

$0.75 \times 2 = 1.5 \quad 1$

$0.5 \times 2 = 1.0 \quad 1$

$(10011.11)_2$

$(19.75)_{10} = (10011.11)_2$

eg2

$$(210.23)_{10} \longrightarrow (?)_8$$

$\div 8 \Rightarrow \underbrace{0 - 7}_{}$ rem

Case-2: Decimal to Octal Base conversion.

210.23

Example: $(\cancel{2010.23})_{10} = (\,)_8$

After Decimal

Before Decimal

Remainder

8	210	2
8	26	2
	3	3
	0	

$0.23 \times 8 = 1.84$ 1

$0.84 \times 8 = 6.72$ 6

$0.72 \times 8 = 5.72$ 5

0.72×8

$(322.165)_8$

$(210.23)_{10} = (322.165)_8$

Challenge

eg3) $(1228.55)_{10} \longrightarrow (?)_{16}$

Topic : Number System

$A \rightarrow 10$
$B \rightarrow 11$
$C \rightarrow 12$

Case-3: Decimal to Hexadecimal Base conversión.

Example: $(1228.55)_{10} = (\ )_{16}$

Before Decimal Remainder

After Decimal

16	1228	12(C)
16	76	12(C)
	4	4
	0	

$0.55 \times 16 = 8.84$ 8

$0.84 \times 16 = 12.8$ 12 (C)

$(4CC.8C)_{16}$

$(1228.56)_{10} = (4CC.8C)_{16}$

[NAT]

#Q. $(93.25)_{10} = (?)_5$

		Rem
5	93	3
5	18	3
	3	3
	0	Stop

$0.25 \times 5 = 1.25$

$0.25 \times 5 = 1.25$

$\underline{(333.11)}_5$

Type 3 : Any base $\longrightarrow$ Any other Base

$$(b1) \underset{T1}{\longrightarrow} (10) \overset{T2}{\longrightarrow} (b2)$$

$$(\quad)_{b1} \longrightarrow (\quad)_{b2}$$

Topic : Number System

How $()_2 \rightarrow ()_{10} \rightarrow ()_8$

Some Special Case $\Longrightarrow$ only for $\boxed{2^x \longleftrightarrow 2^y}$

Case-1: Binary to Octal base conversion

Example: $(10110111)_2 = ()_8$

$()_2 \rightarrow ()_{10} \rightarrow ()_8$

$2^1 \rightarrow 2^3$ ✓

Octal→ manes base 8

2^3

$8 = 2^3$ ✓

Every three digits of binary represent one digit of octal (Right to left)

(010) (110) (111)

2 6 7

Hence, $(10110111)_2 = ()_8 \Rightarrow (267)_8$

$$(101\,1011)_2 \longrightarrow (?)_{16}$$

Topic : Number System

HW

$\boxed{(\ )_2 \rightarrow (\ )_{10} \rightarrow (\ )_{16}}$

$2^1 \rightarrow 2^4$ ✓

Some Special Case

Case-2: Binary to Hexadecimal base conversion

Example: $(1011011)_2 = (\ )_{16}$

Hexadecimal → means base 16

$16 = 2^4$

Every four digits of binary represent one digit of Hexadecimal.

01011011

5 11(B)

Hence, $(1011011)_2 = (\ )_{16}$

$(0101) \rightarrow 5$

$(1011) \rightarrow 11 \rightarrow B$

$\Rightarrow (5B)_{16}$

[NAT]

#Q. $(3221)_4 = (\ )_2$

Appl $(\ )_4 \rightarrow (\ )_{10} \rightarrow (\ )_2 \Rightarrow$ HW

Shortcut $4 = 2^{2}$

$(3\ 2\ 2\ 1)_4$

$(11\ 10\ 10\ 01)_2 \Rightarrow (11101001)_2$

$(3221)_4$

[NAT]

#Q. $(330123)_4 = (\ )_8$

universal

Appr 1: $(\ )_4 \rightarrow \boxed{(\ )_{10}} \rightarrow (\ )_8$

HW ✓

Appr 2: $(\ )_4 \rightarrow \boxed{(\ )_2} \rightarrow (\ )_8$

$$(330123)_4 \longrightarrow (?)_2$$

$$4 = 2^{2}$$

1)

$$\begin{array}{cccccc} 3 & 3 & 0 & 1 & 2 & 3 \\ \downarrow & \downarrow & \downarrow & \downarrow & \downarrow & \downarrow \end{array}$$

$$(\underbrace{111}_{7}\,\underbrace{100}_{4}\,\underbrace{011}_{3}\,\underbrace{011}_{3})_2 \longrightarrow (?)_8$$

$$2 \longrightarrow 2^{3}$$

$$\Rightarrow (7433)_8 \checkmark$$

[NAT]

#Q. $(7634)_8 = (\ )_{16}$

$$(\ )_8 \rightarrow (\ )_2 \longrightarrow (\ )_{16}$$

$$16 = 2^{4}$$

1. $(7634) \rightarrow (111\ 110\ 011\ 100)_2 \Rightarrow (\underbrace{111\,1}\ \underbrace{10\,011}\ \underbrace{1\,100})_2 \rightarrow (\ )_{16}$

$$(1111) \rightarrow F \qquad (1001) \rightarrow 9 \qquad (1100) \rightarrow C$$

$$\Rightarrow (F9C)_{16}$$

[MCQ]

#Q. Find number of solutions of x and y which can satisfy the equation $(43)_8 = (x8)_y$.

HW

A 1

B 2

C 3

D 4

[MCQ]

#Q. The result of addition operation 34 + 43 performed on minimum base is stored in an 8-bit register. The content of register will be

$(34)_5 + (43)_5$

HW

90%

0 1 0 50%

A 01000011

B 00101010

C 01010101

D 01010100

2 mins Summary

$\longrightarrow (\ )_b \longrightarrow (\ )_{10}$

$\longrightarrow (\ )_{10} \longrightarrow (\ )_b$

$\longrightarrow (\ )_{b1} \longrightarrow (\ )_{b2}$

THANK - YOU